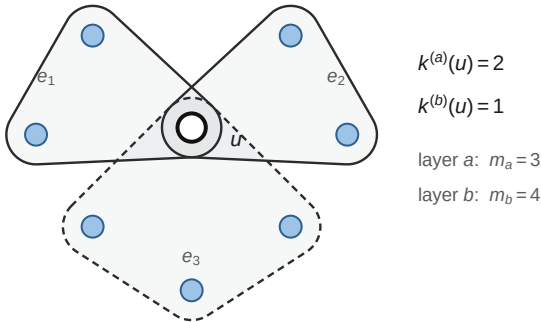


(a) Multiplex hypergraph and hyperdegrees

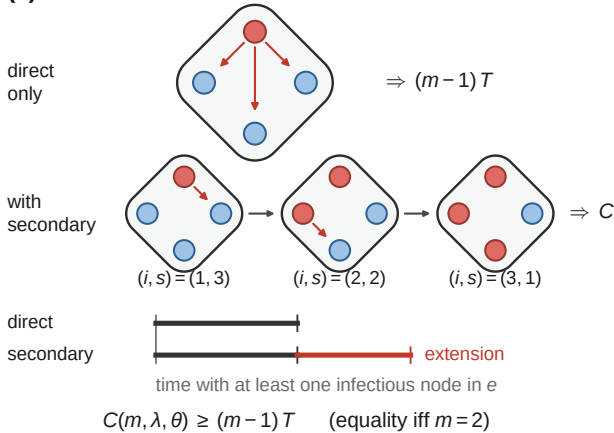


joint hyperdegree distribution $P(\mathbf{k})$, $\mathbf{k} = (k^{(1)}, \dots, k^{(M)})$

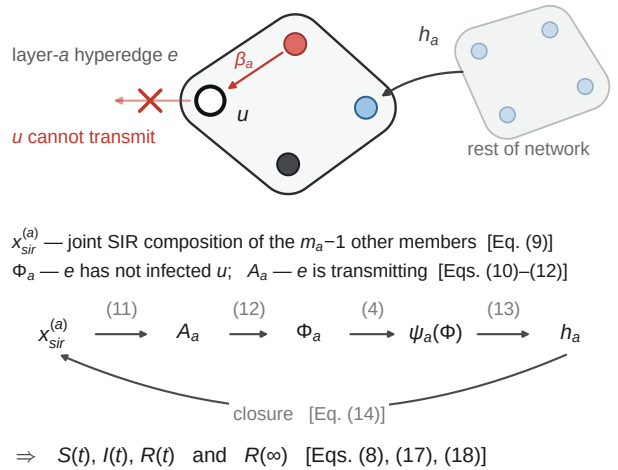
$\Psi(\mathbf{x})$ — uniformly sampled node [Eq. (3)]

$\psi_a(\mathbf{x})$ — node reached via layer a , arrival edge excluded [Eq. (4)]

(c) Rare-infection limit I: within-hyperedge factor C_b



(b) Edge-based closure at finite prevalence



(d) Rare-infection limit II: excess hyperdegree B_{ab}

